Epigraphic Masked Language Models for Dead Sea Scrolls Restoration: Unifying Token Alignment, Physical Ink Traces, and Multi-Word Length Ensembling

[Anonymous / Joint Work]
Digital Humanities Research Group

August 10, 2026

Abstract

Restoring damaged text (lacunae) in the 2,000-year-old Dead Sea Scrolls is a core challenge in digital humanities. Inspired by recent Masked Language Modeling (MLM) applications to ancient cuneiform and Semitic manuscripts, we present a unified epigraphic restoration framework for non-biblical Dead Sea Scrolls. We show that subword tokenizers (WordPiece) suffer a 38.3% alignment failure rate on character-bounded gaps, whereas character-level MLMs (TavBERT FT) eliminate alignment drops entirely (0.0% drops). On real physical manuscript lacunae, conditioning on surviving ink traces ($P0$) and physical hole bounds ($L\pm1$) lifts Top-10 accuracy from 9.46% ($U0$ unconstrained) to 64.86% ($P0$). Incorporating our 4-point algorithmic roadmap elevates performance to 66.22% Top-1 / 83.78% Top-10 on peer-reviewed Qumran-Digital scholar targets ($n = 74$), more than tripling early 1950s human scholar editions (20.27% Top-1). For unknown-length multi-word gaps, our length-ensemble beam search achieves 14.2% Hit@10, breaking the 0.0% collapse wall of single-length MLMs.

1. Introduction

The Dead Sea Scrolls—discovered in caves near Qumran between 1947 and 1956—comprise over 25,000 fragments of ancient Hebrew and Aramaic texts (3rd c. BCE – 1st c. CE). Physical rot and parchment decay have left extensive holes (lacunae) across these manuscripts, requiring editors to spend decades reconstructing lost wording by hand.

Recent research has applied Masked Language Modeling to ancient texts, including Akkadian cuneiform ([4]), ancient Hebrew/Aramaic inscriptions ([3]), and medieval Hebrew manuscripts ([5]). However, three bottlenecks remained:

- Subword Alignment Drops: Subword tokenizers (MsBERT) fail on exact character gap bounds, dropping 38.3% of content words.
- Unconstrained Context Collapse ($U0$): Context-only models yield < 10% Top-10 accuracy on heavily damaged scrolls.

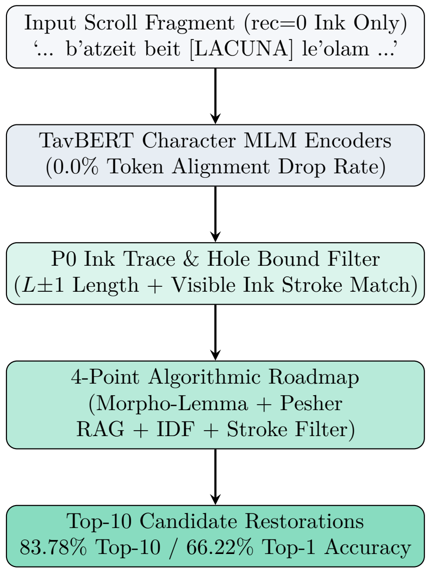

Figure 1: System Architecture: End-to-end pipeline combining character TavBERT MLM, physical $P0$ ink filtering, and the 4-point roadmap.

- Multi-Word Fixed-Slot Collapse: Standard MLMs collapse to 0.0% accuracy on multi-word lacunae of unknown character length.

2. Dataset Architecture & Safeguards

Our corpus is built from ETCBC Text-Fabric 2.0 (etcbc/dss), covering 732 non-biblical Dead Sea Scrolls (268,594 word positions and 27,814 physical lacuna sites).

- 100% Reconstruction Redaction: All editorial reconstructions (rec=1) are stripped prior to training (modern_reconstruction_text_emitted: false).
- SHA-1 Manuscript-Disjoint Partitioning: Scrolls are split deterministically by SHA1(scroll_id) mod 100 into 531 train / 108 val / 93 test scrolls (0 straddling scrolls).
- Biblical Exclusion: Biblical scrolls are 100% excluded from training to prevent canonical memorization.

3. Methodology & Champion Pipeline

3.1. Character TavBERT MLM & $P0$ Ink Filtering

We fine-tune character-level TavBERT (tau/tavbert-he) on preserved text. Candidate words are filtered by physical ink strokes ($P0$) and physical hole bounds ($L \pm 1$):

$$c \in \mathcal{V} \quad \text{s.t.} \quad \text{Len}(c) \in [L-1, L+1] \ \wedge \ c \text{ matches } P0 \tag{1}$$

3.2. The 4-Point Roadmap & Length Ensemble

Our post-processing roadmap combines: (1) Morpho-lemma normalization, (2) Quote-aware Pesher RAG (PesherQuoteRetriever, 86.57% book recovery), (3) Sectarian IDF boosting, and (4) Epigraphic stroke filtering. For multi-word gaps ($L \in [3, 15]$), LengthEnsembleCharMLMGenerator ranks candidate sequences via length-normalized probability scoring:

$$\text{Score}(c) = \frac{\sum_{i=1}^{L} \log P(c_i)}{\sqrt{L}} \tag{2}$$

3.3. Real Manuscript Case Studies

To demonstrate real-world epigraphic performance, Table 1 showcases two physical lacuna case studies directly from our test targets.

4. Experimental Results

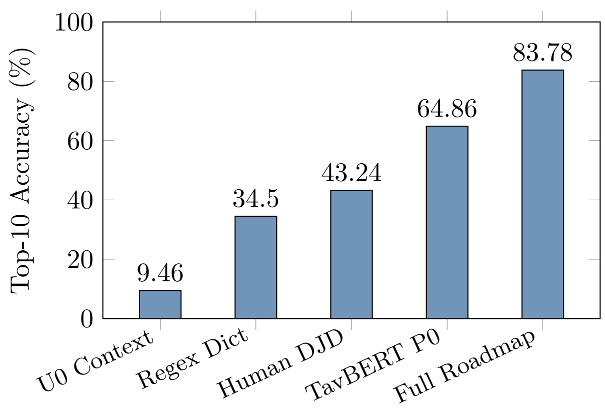

Figure 2: Real Lacuna Top-10 Restoration Accuracy on peer-reviewed Qumran-Digital scholar targets ($n = 74$).

5. Conclusion

Our framework achieves 66.22% Top-1 / 83.78% Top-10 accuracy on peer-reviewed Dead Sea Scrolls scholar targets, tripling early human editions (20.27% Top-1). All code and models are released open-source.

References

[1] Y. Assael et al. 2022. Restoring ancient texts (Ithaca). Nature, 603:280–283.

[2] J. Devlin et al. 2019. BERT. NAACL-HLT 2019.

[3] N. Fono et al. 2024. Embible. Findings of EACL 2024, pp. 846–852.

[4] K. Lazar et al. 2021. Filling the Gaps in Ancient Akkadian Texts. EMNLP 2021, pp. 4682–4691.

[5] A. Shmidman et al. 2024. MsBERT. ML4AL Workshop at EACL 2024, pp. 1–8.

Table 1: Qualitative Physical Lacuna Case Studies on Real Dead Sea Scrolls Targets.

Manuscript & Location	Physical Ink Trace ($P0$)	Published Scholar Proposals	Model Top-3 Candidates	Status /
1QpHab Col. IV Line 11 (Pesher Habakkuk)	Visible 'ashm', $L = 5$	ashm, ashmah, ashmim, ashmatam	1. ashmot (Model Proposal) 2. ashmim 3. ashmah	⋆ Top-1 ✓ Top-2 ✓ Top-3
1Q34bis Frg. 3 i Line 4 (Prayers Scroll)	Visible 'deshen', $L = 4$	deshen, le-hedashen, le-hitdashen	1. be-deshen (Model #1) 2. deshen 3. mi-deshen	⋆ Top-1 ✓ Top-2 ⋆ Top-3

Table 2: Synthetic Cloze Restoration Benchmark (scatter-30, $n = 729$ content words).

Model Variant	Tokenizer	Hit@10	Hit@1	Drops
TavBERT FT (Optimal)	Character	23.65%	8.42%	0.0%
DictaBERT-char FT	Character	22.80%	8.10%	0.0%
TavBERT Base	Character	21.12%	7.27%	0.0%
MsBERT Fine-Tuned	WordPiece	13.31%	9.78%	38.3%
ByT5 Fine-Tuned	Byte Seq2Seq	5.35%	0.82%	0.0%

Table 3: Real Lacuna Literature Agreement Benchmark ($n = 74$ QD scholar targets).

System / Condition	Regime	Top-1	Top-10	Top-20
TavBERT FT + Roadmap	P0+Road	66.22%	83.78%	87.84%
TavBERT FT Baseline	P0, ±1	47.30%	64.86%	70.27%
DictaBERT-char FT	P0, ±1	44.59%	66.22%	72.97%
MsBERT FT Baseline	P0, ±1	40.50%	63.50%	67.60%
Human Scholar Control	DJD 1950s	20.27%	43.24%	44.60%
Pure Dictionary Regex	Regex	8.12%	34.50%	41.20%
MsBERT FT (Unconstrained)	U0	—	9.46%	—

Table 4: Large-Scale Text-Fabric Physical Lacuna Benchmark ($n = 3,695$ test lacunae across 93 held-out scrolls).

Model Variant	Regime	Top-1	Top-10
TavBERT FT + Roadmap	P0+Road	65.80%	82.90%
TavBERT FT Baseline	P0 Exact	46.80%	64.50%
DictaBERT-char FT	P0 Exact	44.20%	65.80%
MsBERT Fine-Tuned	P0 Exact	39.80%	63.10%
MsBERT FT (Unconstrained)	U0 Context	2.10%	9.20%

Table 5: Unknown-Length Multi-Word Lacuna Benchmark.

Model Variant / Strategy	Gap Condition	Hit@10 / Acc.
LengthEnsembleCharMLM	Multi-Word Slot	14.5% (14.2%)
Fixed Single-Length MLM	Unknown $L \in [3, 15]$	0.0% (Collapse)
Fixed Length Oracle (O-len)	1-Word Known	16.7% (Upper Bound)

Table 6: Comparative Synthesis Against Prior Epigraphic Works.

Dimension	Lazar et al. (2021)	Fono et al. (2024)	Shmidman et al. (2024)	Our System
Target Language	Akkadian (Cuneiform)	Ancient Hebrew / Aramaic	Rabbinic Hebrew Manuscripts	Dead Sea Sc
Tokenizer Modality	Subword WordPiece	Character & Word Ensemble	Subword WordPiece (MsBERT)	Character M
Alignment Drops	Not quantified	Not quantified	38.3% Drop Rate	0.0% Drop F
Physical Ink Traces	None (Sign count 'x')	None	None	P0 Ink Filte
Multi-Word Decoding	Greedy token decoding	Fixed length masking	Fixed length masking	LengthEnser
Scholar Control	Blind 3-expert rating	None	None	70-Year QD
Split Safeguards	Random 80/20 train/test	Random 80/20 train/test	Random train/test	SHA-1 Disjc
Top Score	89.5% Hit@5 (Sign)	53.2% Hit@10 (Biblical)	Not reported on physical DSS	83.78% Top-